

ADVANCED DATA VISUALIZATION TECHNIQUES

Smart City Traffic and Accident Analytics

COURSE PROJECT REPORT

Submitted for the Course Project MTech in MID

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0. ABSTRACT

Urban traffic management and road safety have become important challenges because cities experience increasing vehicle movement, changing road conditions, congestion, and traffic accidents. Large volumes of accident records contain useful information about accident locations, severity, weather conditions, road characteristics, and time-based patterns. However, examining these records manually makes it difficult to identify trends, compare locations, and communicate useful findings efficiently.

The Smart City Traffic and Accident Analytics Platform is designed to organize traffic accident data and present meaningful information through interactive visualizations and machine-learning techniques. The system uses a historical road-accident dataset containing attributes such as accident location, date, time, road type, weather, traffic density, accident cause, accident severity, and casualties. The data preparation process includes checking data quality, handling missing values, removing or investigating duplicates, transforming relevant fields, and preparing analytical variables.

The visualization component provides an interactive dashboard containing summary indicators, accident severity charts, city-wise and state-wise comparisons, hourly and daily accident trends, weather-based analysis, traffic-density comparisons, and geographic accident maps. Users can explore patterns through filters and interactive chart elements rather than relying only on static reports.

The machine-learning component uses a classification approach, such as Random Forest, to investigate accident severity patterns using available input features. Model performance is evaluated using appropriate measures, including accuracy, precision, recall, F1-score, and a confusion matrix. Feature importance can also be examined to understand which variables contribute to model predictions. The predictions are treated as analytical estimates and are not presented as guaranteed future outcomes.

The platform is intended to support exploratory analysis, data-driven decision-making, and road-safety research. By combining data preprocessing, interactive visualization, geographic exploration, and machine learning within one analytical workflow, the project demonstrates how data science techniques can make traffic accident information easier to interpret. The reliability of its findings depends on dataset quality, model evaluation, and the representativeness of the available observations.

Keywords: Smart city, traffic analytics, road accidents, advanced data visualization, machine learning, Random Forest, accident severity, interactive dashboard, geographic analysis, road safety.

1. INTRODUCTION

1.1 Background and Motivation

Smart cities depend on the effective use of information technology, data analytics, and intelligent transportation systems to improve urban services. Traffic management is an important part of this development because road accidents, congestion, and unsafe driving conditions can affect public safety, travel efficiency, and the overall quality of urban life.

Modern transportation systems generate different types of data, including accident reports, geographic coordinates, road characteristics, traffic conditions, weather information, and time-based observations. When these records are organized and analyzed correctly, they can help reveal recurring accident patterns and differences between locations.

Traditional accident analysis often relies on spreadsheets, static charts, and manual examination of individual records. These approaches may be sufficient for small datasets but can become difficult to use when the number of records and attributes increases. Static reports also limit the ability to investigate relationships between variables or compare multiple categories interactively.

The proposed Smart City Traffic and Accident Analytics Platform addresses these challenges through advanced interactive data visualization and machine learning. It converts accident records into graphical summaries, interactive charts, geographic views, and analytical insights.

The platform can help users examine questions such as:

- Which cities or regions contain the highest number of recorded accidents?
- How are accidents distributed across different severity categories?
- How do recorded accidents vary by hour, day, or month?
- How do accident records differ across weather conditions and road types?
- Which available features contribute to accident-severity classification?
- How can accident information be presented clearly for further investigation?

By combining data visualization with machine learning, the project provides a structured approach to understanding historical road-accident data.

1.2 Problem Statement

Road accident datasets contain valuable information, but identifying meaningful patterns across large numbers of records can be difficult using manual analysis alone. Accident

information may be distributed across multiple columns, including location, time, road type, weather, traffic density, accident cause, severity, and casualty counts.

Without a unified analytical interface, users may find it difficult to compare accident distributions, explore geographic patterns, identify temporal trends, and evaluate relationships between different variables. Furthermore, descriptive statistics alone cannot automatically classify accident severity or reveal which features influence a trained classification model.

Therefore, the project aims to develop an integrated platform that preprocesses historical accident data, generates interactive visualizations, supports geographic exploration, and applies machine-learning classification to investigate accident severity.

The system is designed as an analytical support tool. Its outputs must be interpreted according to the quality of the dataset and the measured performance of the implemented models.

1.3 Scope and Objectives

- Import historical road-accident records from a structured dataset.
- Inspect the dataset for missing values, duplicate records, inconsistent categories, and invalid values.
- Prepare the dataset for descriptive analysis and machine learning.
- Create interactive charts for accident severity, location, time, weather, and road conditions.
- Develop a geographic view of accident locations when valid coordinate data are available.
- Implement a machine-learning classification model for accident severity.
- Evaluate the classification model using suitable performance metrics.
- Display summary indicators, charts, and analytical outputs through a unified dashboard.
- Provide data filtering and record-exploration capabilities.
- Document the results, limitations, and potential future improvements.

1.4 Organization of the Report

This report is divided into nine chapters. Chapter 1 introduces the project and its objectives. Chapter 2 describes the project goals, related work, research gap, and key contributions. Chapter 3 presents technical requirements and feasibility. Chapter 4 explains the architecture and design. Chapter 5 describes the data-processing, visualization, machine-learning, and

testing methodology. Chapter 6 presents the dashboard modules. Chapter 7 discusses results and model evaluation. Chapter 8 summarizes the project and proposes future improvements. Chapter 9 lists the technical and research references.

CHAPTER 2 — PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

2.1.1 Data visualization in transportation analytics

Data visualization converts numerical and categorical information into visual representations that help users identify patterns, compare values, and communicate findings. Bar charts, line charts, scatter plots, heatmaps, and geographic maps are commonly used to examine transportation-related data.

In traffic accident analysis, visualization can show how recorded incidents vary by location, severity, time, weather, or road type. Interactive visualizations extend this capability by allowing users to select categories, apply filters, inspect individual values, and compare subsets of data.

The R visualization ecosystem, particularly ggplot2 and Plotly, provides tools for constructing analytical graphics. JavaScript libraries such as Chart.js can also support interactive dashboards in web applications.

2.1.2 Machine learning for accident analysis

Machine learning can identify patterns in historical records and learn relationships between selected input variables and a target variable. Classification methods are useful when the target is categorical, such as minor, major, or fatal accident severity.

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. It can model nonlinear relationships and interactions among input features. Its performance can be compared with alternative classifiers using a consistent training and testing procedure.

However, machine-learning predictions depend on the quality and representativeness of the training data. Data leakage, class imbalance, missing values, and inconsistent labels can produce misleading evaluation results.

2.1.3 Geographic visualization

Geographic visualization places observations on a map using latitude and longitude or other location information. Mapping accident records can reveal spatial concentrations in the available data and make geographic comparisons easier.

Interactive mapping tools such as Leaflet support markers, zooming, popups, and map layers. When accident coordinates are available and sufficiently accurate, the platform can display recorded incident locations and support location-based exploration.

A concentration of recorded incidents should not automatically be interpreted as a high probability of future accidents. Such conclusions require additional information about traffic exposure, road conditions, reporting coverage, and other relevant factors.

2.1.4 Interactive analytical dashboards

An analytical dashboard combines multiple indicators and visualizations in one interface. Summary cards provide a quick overview, while charts, maps, filters, and tables allow detailed investigation.

A well-designed dashboard should provide readable labels, consistent units, clear navigation, informative tooltips, and explanations of calculated indicators. These features help users understand not only the displayed values but also their meaning and limitations.

2.2 Research Gap

Many basic accident-analysis implementations focus on only one analytical task, such as plotting accident counts or training a classification model. Separate tools can make it harder for users to move between overall statistics, geographic patterns, temporal analysis, and model outputs.

The proposed project addresses this integration challenge by combining several analytical components within one platform:

- Data preparation and validation.
- Interactive accident visualization.
- Geographic accident exploration.
- Machine-learning-based severity classification.
- Model performance and feature-importance analysis.
- Filterable records and dashboard-level summaries.

The research contribution is the integration of these techniques into an accessible analytical workflow. The project does not claim to invent a new visualization method or machine-learning algorithm.

2.3 Novelty and Key Contributions

The platform's contribution lies in connecting accident data preparation, interactive exploration, geographic visualization, and classification results.

1. Unified accident analytics: The dashboard combines accident statistics and multiple analytical views instead of requiring users to inspect separate reports.

2. Interactive visual exploration: Users can investigate accident distributions through charts, filters, and geographic views.

3. Machine-learning integration: A classification model can be used to analyze accident-severity categories and display its measured performance.

4. Feature-importance analysis: The platform can present model-derived feature importance to help users examine which input variables contribute to predictions. Feature importance indicates model behavior, not proof of causation.

5. Data transparency: A data-explorer module allows users to inspect the records underlying visualizations, making it easier to verify displayed summaries.

6. Modular architecture: Data preparation, model training, visualization, and dashboard presentation can be developed and tested independently.

7. Responsible interpretation: The system distinguishes recorded accident patterns, model predictions, and actual future accident probabilities.

2.4 Project Objectives

The major objectives are:

- To develop a centralized dashboard for road-accident analysis.
- To transform raw accident records into a consistent analytical dataset.
- To visualize severity distributions and geographic patterns.
- To investigate temporal trends and relationships between recorded accident attributes.
- To implement and evaluate a machine-learning classification model.
- To provide interactive data filtering and detailed record inspection.
- To improve the accessibility and interpretability of accident information.
- To establish a foundation for future predictive and intelligent transportation features.

2.5 Project Plan

The project can be developed in six phases.

Phase	**Activities**	**Deliverable**
1. Data Preparation	Load, inspect, clean, and validate the dataset	Prepared CSV
2. Exploratory Analysis	Generate summaries and examine variable distributions.	Analytical summaries
3. Visualization	Create charts, maps, and interactive filters.	Visualization modules
4. Machine Learning	Train and evaluate the classifier.	Model and evaluation outputs
5. Dashboard Integration	Connect the data, charts, and model outputs.	Integrated platform
6. Testing and Documentation	Verify outputs, capture screenshots, and prepare the report.	Tested project and report

2.6 Expected Outcomes

The expected outcome is an interactive analytical platform that helps users understand historical accident information without manually inspecting every record.

The completed implementation should provide:

- A summary of accident records and related indicators.
- Interactive severity and location comparisons.
- Temporal and weather-based charts.
- A geographic accident map, when valid coordinates exist.
- A machine-learning classification workflow.
- Model evaluation metrics and feature-importance visualization.
- A filterable data table.
- A documented testing process and explanation of limitations.

These are intended project outcomes. Any claim that a feature has been implemented or achieved should be supported by the actual running application and its outputs.

CHAPTER 3 — TECHNICAL SPECIFICATION

3.1 Functional Requirements

ID	Requirement
FR-01	Load historical accident records and inspect the dataset structure.
FR-02	Generate descriptive summaries and visualizations for selected accident attributes.
FR-03	Train and evaluate severity-classification baselines.
FR-04	Read a traffic video and process frames with the configured object detector.
FR-05	Track detections and count cars, motorcycles, buses, and trucks.
FR-06	Calculate per-frame vehicle totals, traffic-level labels, and a bounded heuristic score.
FR-07	Export frame-level results to CSV and save an annotated video.
FR-08	Display analytical outputs through an R Shiny dashboard.

3.2 Non-Functional Requirements

Requirement	Description
Reliability	Validate required files, columns, output paths, and count arithmetic.
Performance	Record processing throughput and hardware; do not equate throughput with accuracy.
Usability	Provide readable chart labels, filters, explanatory captions, and consistent navigation.
Reproducibility	Record data version, model weights, confidence threshold, tracker configuration, and software versions.
Privacy	Restrict access to sensitive data and avoid exposing identifying video details unnecessarily.
Maintainability	Organize preprocessing, model execution, dashboard logic, and output files into separate modules.

3.3 Feasibility Study

3.3.1 Technical feasibility

The project can be implemented using established data-processing, visualization, and machine-learning libraries. R can be used for data cleaning and analysis, while HTML, CSS, and JavaScript can provide a browser-based dashboard. A backend service can supply the processed data and analytical results to the frontend.

The system is technically feasible as an academic prototype using a local computer and a structured dataset. Actual performance depends on dataset size, implementation choices, hardware, and the complexity of model training.

3.3.2 Economic feasibility

The core project can use open-source tools, reducing software licensing costs. A local implementation can operate without a paid cloud service.

Potential costs may arise if the project is deployed on a hosted server, uses commercial map services, or requires additional infrastructure. No deployment-cost estimate is claimed here.

3.3.3 Operational feasibility

The dashboard is designed to be used by students, researchers, and analysts who need to explore accident records. Interactive charts and summary indicators can make the analysis easier to understand than a collection of raw spreadsheets.

Users should still be able to inspect the underlying records and methodology rather than treating a chart or machine-learning prediction as conclusive evidence.

3.3.4 Social and ethical feasibility

Road-safety analytics may help identify areas for further investigation, but inaccurate data or model predictions could lead to misleading conclusions. Geographic concentrations should be interpreted in the context of data coverage and traffic exposure.

The project should avoid exposing personally identifiable information and should not be used as the sole basis for enforcement, emergency response, or decisions affecting individuals.

3.4 System Specification

The system uses two input paths: a tabular road-accident dataset for historical analysis and an MP4 traffic video for vehicle-level analysis. Python and its computer-vision libraries support frame processing, YOLO11n inference, ByteTrack tracking, and CSV/video generation. R

Shiny provides the interactive presentation layer for charts, maps, summary cards, and data exploration. The components exchange files rather than requiring a single real-time service, which keeps the prototype straightforward to run and inspect. Exact package versions, input files, model configuration, and output paths should be recorded when reproducing an experiment.

Component	Technology / Configuration
Historical data	Indian road-accident dataset; project documentation reports 20,000 records and 24 features.
Video input	Traffic_Video.mp4; reported resolution 1920 × 1080 and nominal 59.94 FPS.
Object detector	Ultralytics YOLO11n; inference image size 640; confidence threshold 0.20.
Tracking	ByteTrack using bytetrack.yaml; persistent tracking enabled in the v2 script.
Video handling	Python and OpenCV.
Modeling	Random Forest and multinomial logistic regression baselines.
Dashboard	R Shiny with data visualization and geographic views.
Output artifacts	Frame-level CSV, annotated MP4, plots, and dashboard views.

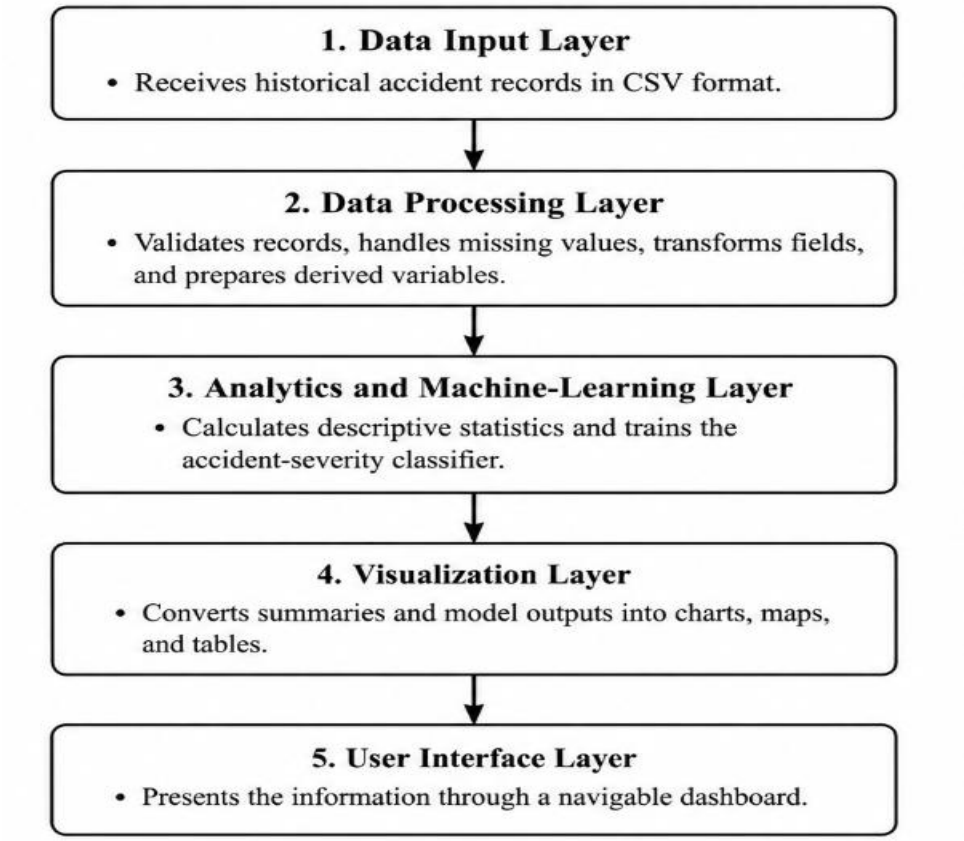
CHAPTER 4 — DESIGN APPROACH AND DETAILS

4.1 System Architecture

The system architecture consists of five major layers:

1. Data input layer: Receives historical accident records in CSV format.
2. Data processing layer: Validates records, handles missing values, transforms fields, and prepares derived variables.
3. Analytics and machine-learning layer: Calculates descriptive statistics and trains the accident-severity classifier.
4. Visualization layer: Converts summaries and model outputs into charts, maps, and tables.
5. User interface layer: Presents the information through a navigable dashboard.

The overall system can be represented as follows.

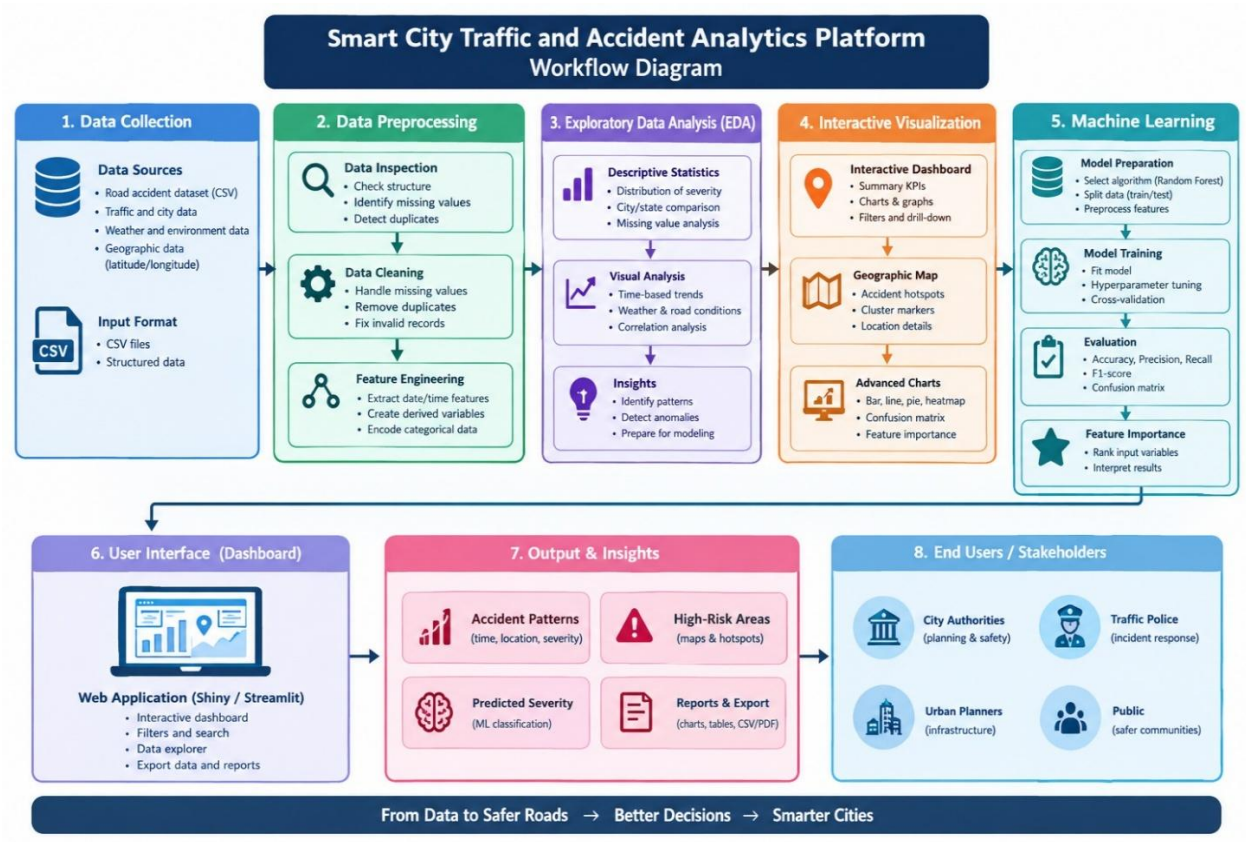


4.2 Data Flow Diagram

The data flow begins when the system reads the accident dataset. The dataset is validated and prepared before being passed to two analytical branches.

The descriptive branch generates summaries by severity, location, time, and other available attributes. The machine-learning branch prepares the target variable, splits the data into training and testing subsets, trains the classifier, and generates evaluation metrics.

The dashboard combines the available outputs and presents them through an interactive interface.

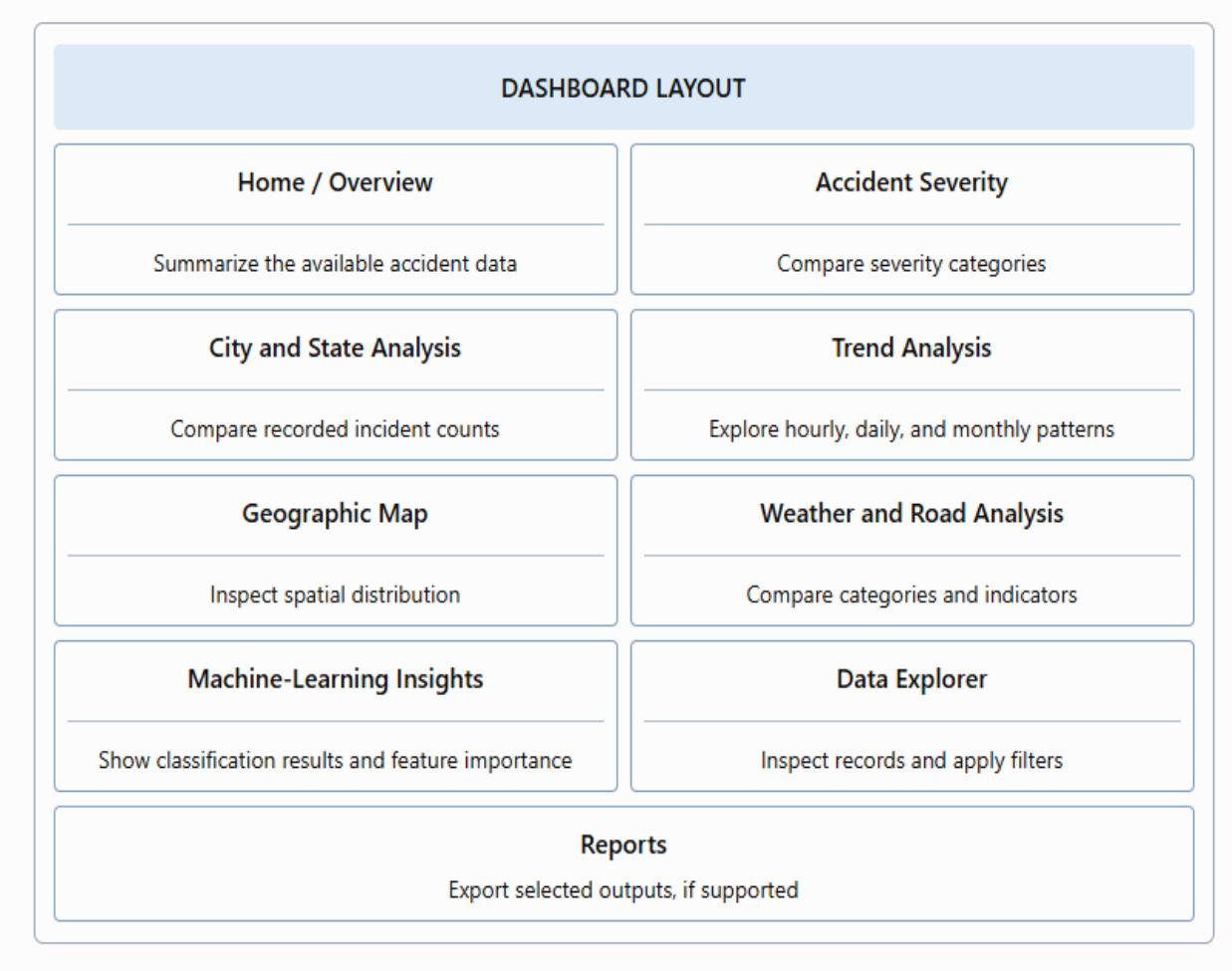


4.3 Dashboard Design

The dashboard is organized into analytical modules so that users can move from a high-level overview to detailed investigation.

The executive overview provides summary indicators and selected charts. The geographic module displays incident locations when suitable coordinates exist. The trend module examines accident distributions over time. The risk-analysis module compares recorded indicators across available categories. The machine-learning module presents classification outputs and evaluation metrics. The data explorer displays filtered records.

The interface should maintain consistent typography, spacing, chart labels, and navigation. Colors should be used consistently to distinguish severity categories, but color alone should not be the only way information is communicated.



4.4 Machine-Learning Architecture

The machine-learning component uses accident records to train a classification model. The target variable is the accident-severity category. Candidate input variables may include weather, road type, traffic density, time, visibility, and other fields available in the dataset.

The architecture consists of the following stages:

1. Select relevant input features and the target variable.
2. Validate data types and missing values.
3. Convert categorical variables into a suitable model representation.
4. Split the dataset into training and testing subsets.
5. Train the Random Forest classifier.
6. Generate predictions for the held-out test subset.
7. Calculate evaluation metrics.

8. Display the evaluation outputs in the dashboard.

The train-test split must be performed before any operation that learns patterns from the data, and preprocessing must avoid leaking information from the test set. If records are grouped by location or time, an appropriate split strategy should also be considered.

4.5 Data Flow and Processing

The platform follows a repeatable processing sequence:

- Load the original accident CSV.
- Verify that the expected columns exist.
- Examine missingness, duplicate records, and invalid values.
- Standardize data types and category labels.
- Create time-derived fields where appropriate.
- Generate descriptive summaries.
- Produce visualization-ready data.
- Train and evaluate the classification model.
- Save or serve the analytical outputs.
- Display the results in the dashboard.
- Verify the dashboard values against the underlying data.

This separation improves maintainability because changes to the dataset-processing logic do not need to require a complete redesign of the frontend.

CHAPTER 5 — METHODOLOGY AND TESTING

5.1 Dataset Preparation

The dataset is the foundation of the platform. The project uses a historical Indian road-accident dataset in CSV format. The version associated with the existing Smart City project has been described as containing 20,000 records and 24 original features. These figures should be reconfirmed against the exact CSV used for the final experiment.

The available fields include the following categories.

Attribute Group	Example Fields	Purpose
Identification	accident_id	Identify individual records
Geographic	city, state, latitude, longitude	Analyze accident locations
Temporal	date, time, hour, day_of_week	Examine accident timing
Road Conditions	road_type, lanes, traffic_signal	Compare road-related categories
Environment	weather, visibility, temperature	Examine environmental conditions
Traffic	traffic_density, is_peak_hour	Compare traffic-related indicators
Accident Details	cause, accident_severity	Analyze recorded causes and severity
Impact	vehicles_involved, casualties	Summarize recorded accident impact
Derived Fields	is_weekend, risk_category	Support selected analytical comparisons

The actual column names may vary depending on the dataset version.

5.1.1 Data inspection

The first step is to inspect the structure of the CSV file. This includes the number of rows and columns, variable types, unique category values, missing values, and possible duplicate records.

The inspection helps identify fields that need transformation or further validation before they are used in charts or machine learning.

5.1.2 Handling missing values

Missing values must be evaluated according to the meaning of each column. Numerical fields may require suitable imputation or exclusion, while categorical fields may require an explicit unknown category or another justified strategy.

A missing value must not automatically be replaced with zero because zero may represent a meaningful measurement. A field with extensive missingness should be examined carefully before being included as a predictor.

5.1.3 Duplicate and invalid records

Duplicate records can inflate accident counts if they represent repeated copies of the same incident. However, two records with similar attributes do not necessarily describe the same accident.

Potential duplicates should be identified using appropriate record identifiers or a justified combination of fields. Invalid coordinates, impossible dates, inconsistent category labels, and out-of-range numerical values should also be investigated.

5.1.4 Feature engineering

Feature engineering creates useful analytical variables from existing fields. Depending on the available data, examples include:

- Extracting hour, weekday, month, and year from a date-time field.
- Creating a weekend indicator.
- Grouping accident records by location and severity.
- Creating hourly or monthly accident summaries.
- Standardizing severity labels.
- Preparing categorical variables for machine-learning algorithms.

Derived variables should be documented, and their calculations should be consistent across the training data and any later predictions.

5.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the structure and distribution of the dataset before building the final dashboard or model.

5.2.1 Accident severity analysis

A frequency table and bar chart can show the number of records in each severity category. This helps reveal whether the target classes are balanced or whether some classes occur much less frequently than others.

A class imbalance is important because a model can achieve high overall accuracy while performing poorly on a less common class.

5.2.2 City-wise and state-wise analysis

Accident records can be grouped by city or state to compare the number of recorded incidents in different locations.

These comparisons describe the dataset. They do not automatically establish which location is the most dangerous because population, traffic volume, road length, reporting coverage, and other exposure factors may differ.

5.2.3 Time-based analysis

Hourly, daily, monthly, and yearly summaries can reveal patterns in the timing of recorded accidents.

A line chart is suitable for examining trends across ordered time periods. A heatmap can show the relationship between hour of day and day of week, provided that the aggregation and axis labels are clear.

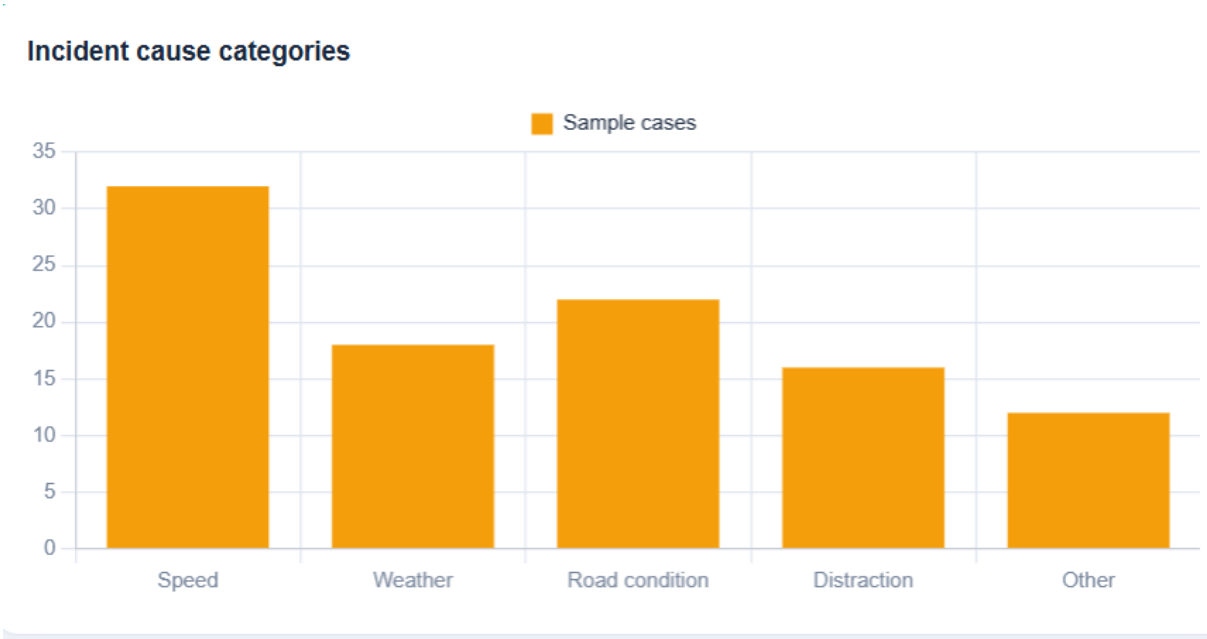
5.2.4 Weather and road-condition analysis

The project can compare accident counts across weather conditions, visibility categories, road types, traffic density, and other available fields.

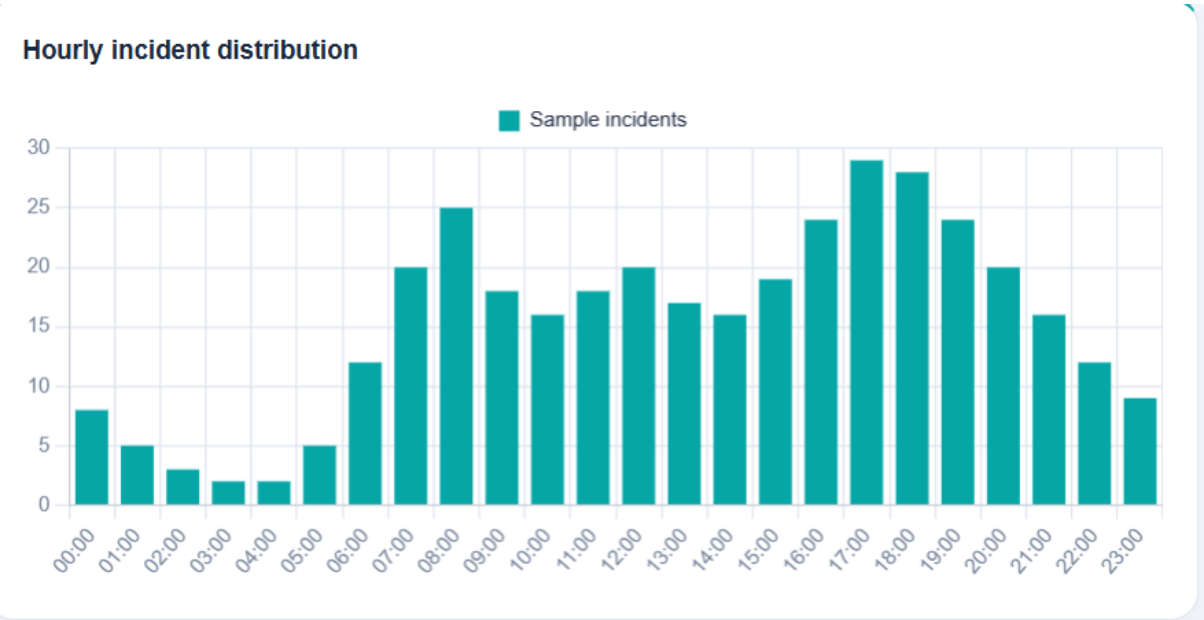
Where appropriate, the analysis can compare severity distributions within each category. The number of recorded accidents should be distinguished from a normalized rate or a calibrated risk estimate.

5.3 Advanced Data Visualization

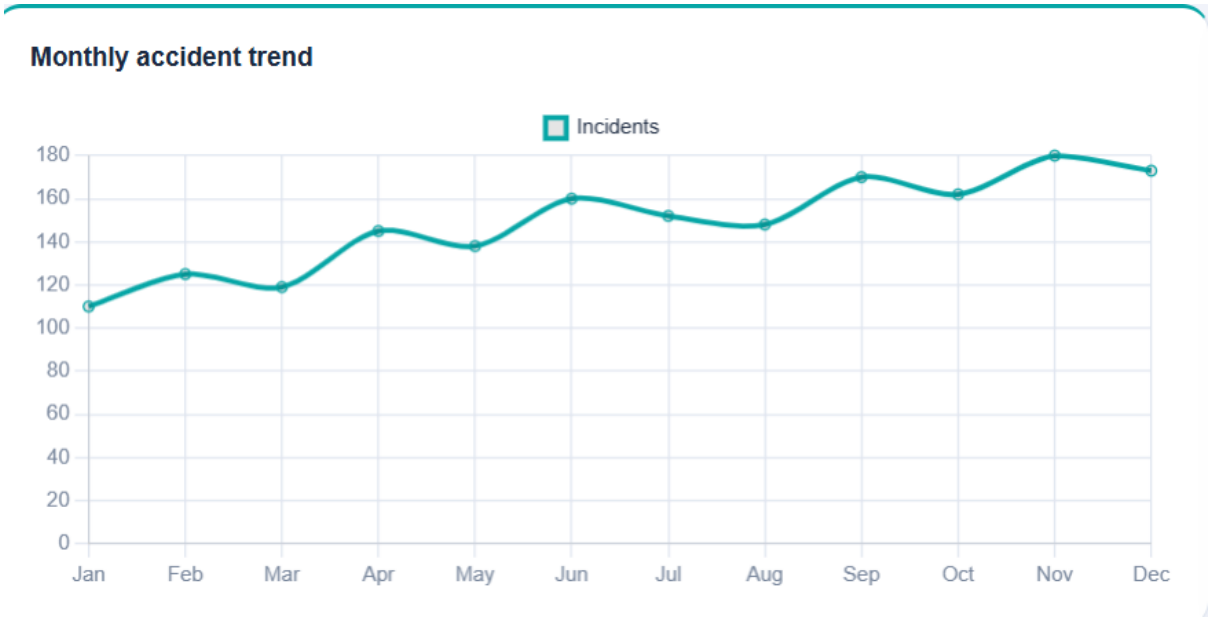
Advanced visualization is central to the proposed platform because it transforms complex data into accessible analytical views.



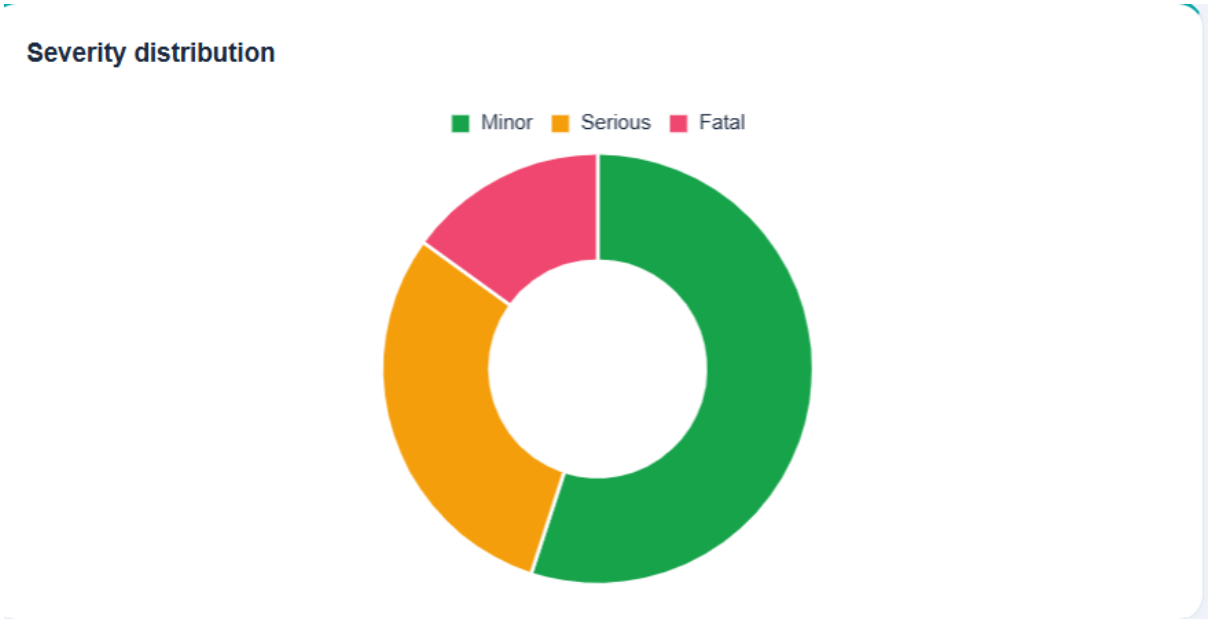
Compares the number of recorded accidents across severity categories, cities, states, road types, and weather groups.



Displays changes in accident counts over time, helping users identify increases, decreases, and recurring patterns.

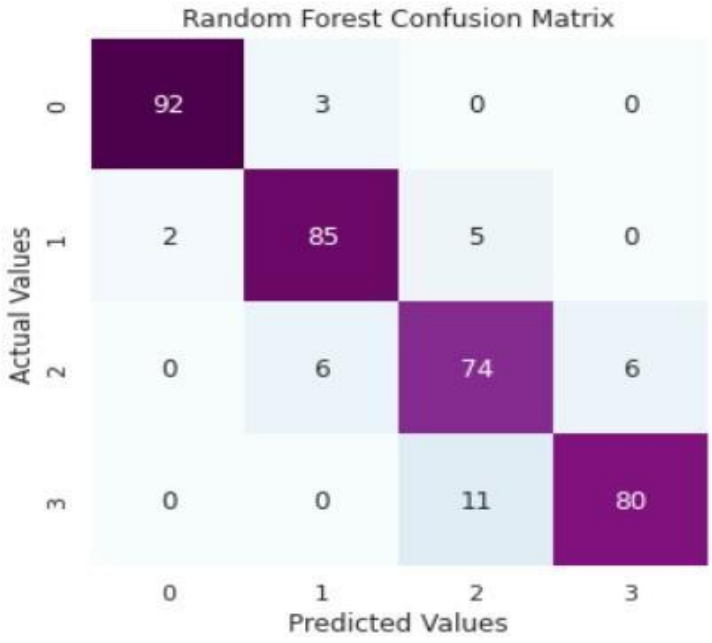


Shows the proportion of records belonging to different severity categories. It is most useful when the number of categories is limited.



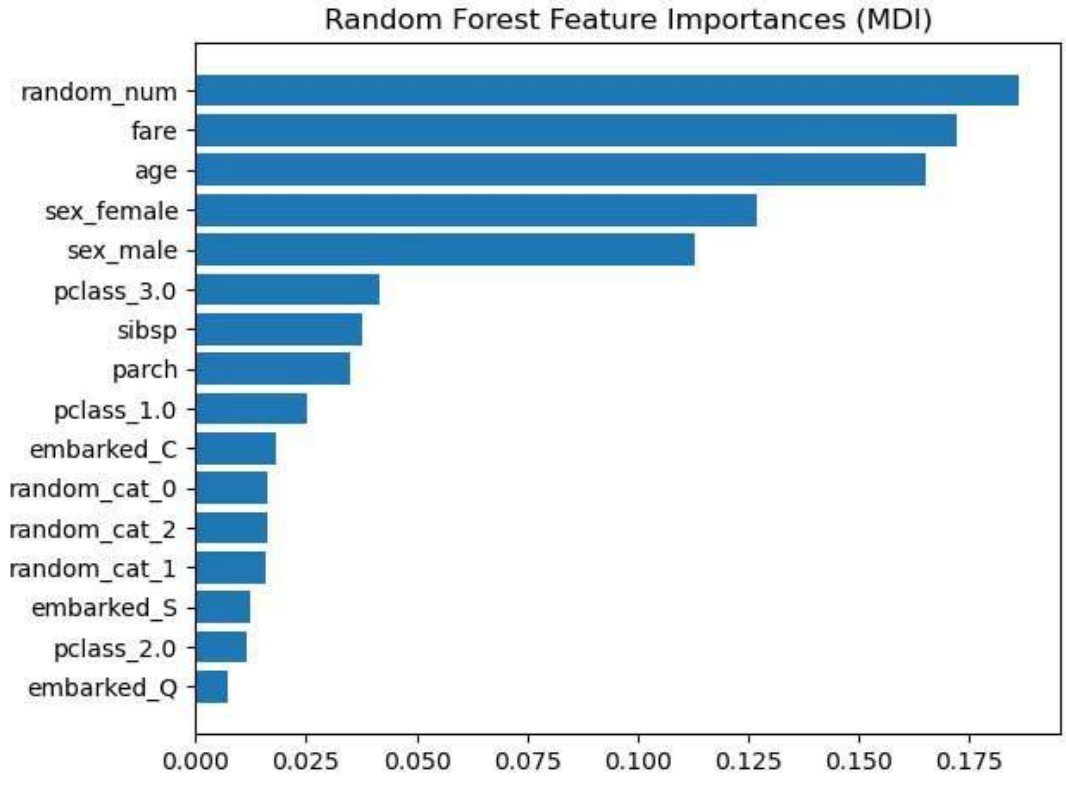
Geographic map

Displays accident locations or aggregated geographic concentrations. A heatmap can show clusters, while individual markers allow record-level inspection.



Confusion matrix

Compares actual severity labels with model predictions to reveal correct classifications and the categories most frequently confused.



Feature-importance chart

Ranks input variables according to a selected model-importance measure. It helps interpret model behaviour but does not prove that a feature causes accidents.

5.3.7 Interactive filtering

The dashboard can include filters for city, severity, date range, weather, and road type. When a filter changes, the associated charts and tables should update consistently.

Tooltips should display meaningful values, including category labels, record counts, and percentages where applicable. Axis labels, units, and titles should remain readable.

5.3.8 Geographic visualization

Where latitude and longitude are available, a mapping library such as Leaflet can display accident markers. Marker popups may show non-sensitive information such as severity, date, or road category.

If the source does not contain reliable geographic coordinates, the map should not fabricate accident locations. A city-level summary or state-level visualization is preferable in that situation.

5.4 Machine-Learning Methodology

5.4.1 Selection of the model

Random Forest is a suitable baseline for a tabular classification problem because it combines predictions from multiple decision trees and can model nonlinear relationships.

For this project, the target variable is accident severity, assuming that the dataset contains reliable severity labels. Candidate predictors can include road type, weather, visibility, traffic density, time-related variables, and other appropriate fields.

Variables that directly encode the target or were derived from the outcome must be excluded if they would cause data leakage.

5.4.2 Data splitting

The dataset should be divided into training and testing subsets. An 80:20 split is one possible choice, provided that the dataset size and evaluation requirements support it.

For example, if there are 20,000 usable records and an 80:20 split is applied after data validation, the split would contain 16,000 training records and 4,000 test records.

This is an illustrative calculation, not a claim that the final dataset has exactly that number of usable records. If the classes are imbalanced, stratified splitting can help preserve class proportions.

5.4.3 Model training

The training subset is used to fit the Random Forest classifier. Model settings, including the number of trees and other relevant parameters, should be recorded so the experiment can be reproduced.

Any parameter tuning must use only the training data, typically through cross-validation. The test subset should remain separate until the final evaluation.

5.4.4 Prediction and evaluation

After training, the model generates severity predictions for the held-out test data. The predictions are compared with the known labels to calculate the classification metrics.

Accuracy

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Here, (TP) denotes True Positives, (TN) denotes True Negatives, (FP) denotes False Positives, and (FN) denotes False Negatives for a particular class.

For a multiclass problem, precision, recall, and F1-score should be reported per class alongside macro-averaged and weighted-averaged measures when appropriate.

5.4.5 Feature importance

The model can generate feature-importance values to show which variables contributed most according to the selected importance method.

The dashboard can display these values as a horizontal bar chart. The report should explain which importance measure was used and avoid treating the ranking as evidence of causal relationships.

5.4.6 Model limitations

The model may be affected by class imbalance, incomplete records, label quality, feature leakage, and differences between the dataset and real-world conditions.

Consequently, model predictions should be presented as classifications based on the available data. A severity classifier is not automatically a model that predicts the location or time of future accidents.

5.5 Testing Strategy

Testing verifies that the platform processes data correctly and displays consistent outputs.

Test ID	Test case	Expected result
TC-01	Load a valid CSV	The dataset loads successfully.
TC-02	Load a missing or invalid file	A clear error is displayed.
TC-03	Inspect required columns	Missing required fields are identified.
TC-04	Process missing values	The configured missing-data strategy is applied.
TC-05	Calculate accident counts	Summary values match the underlying records.
TC-06	Apply dashboard filters	Relevant charts and tables update consistently.
TC-07	Generate a geographic map	Valid coordinate records are mapped correctly.
TC-08	Train the classifier	Training completes or returns a meaningful error.
TC-09	Evaluate predictions	Metrics are calculated from held-out predictions.
TC-10	Inspect confusion matrix	Class labels and counts are aligned correctly.
TC-11	Review feature importance	Features and importance values are displayed correctly.
TC-12	Check the data explorer	Displayed records match the active filters.

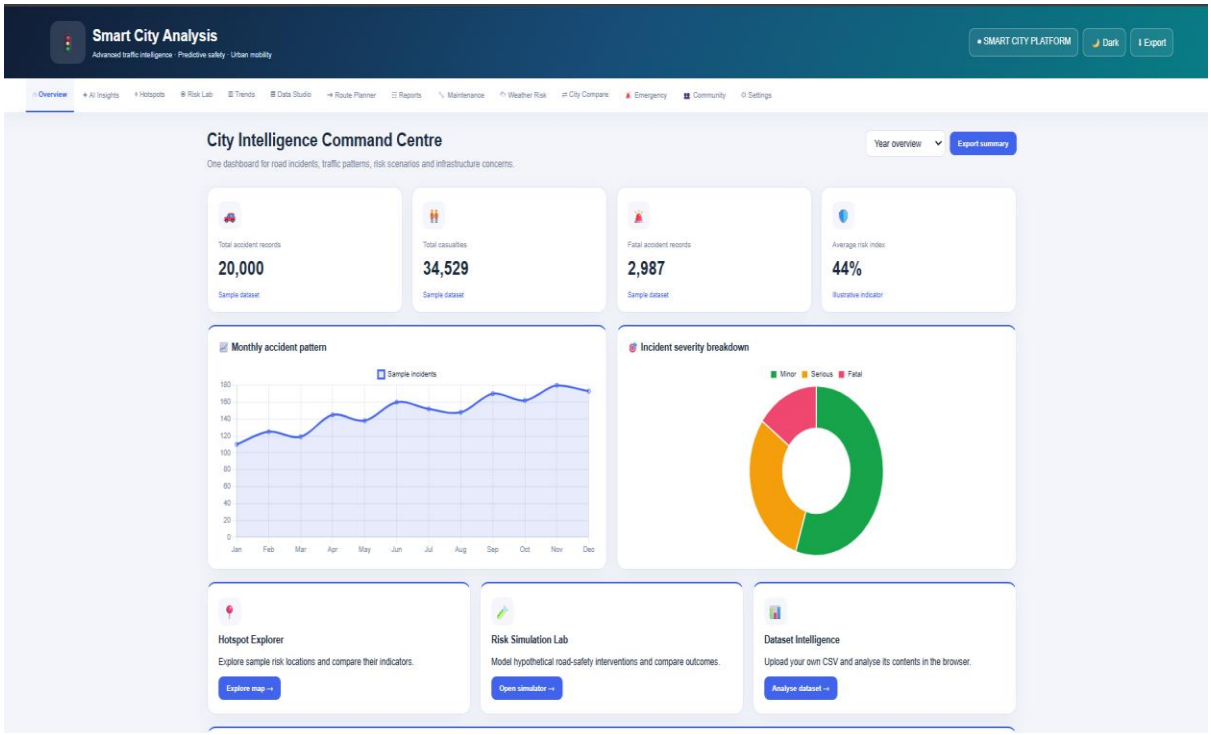
TC-13	Check browser layout	Navigation and charts remain usable at supported screen sizes.
TC-14	Verify exported output	The exported file contains the expected fields and records.

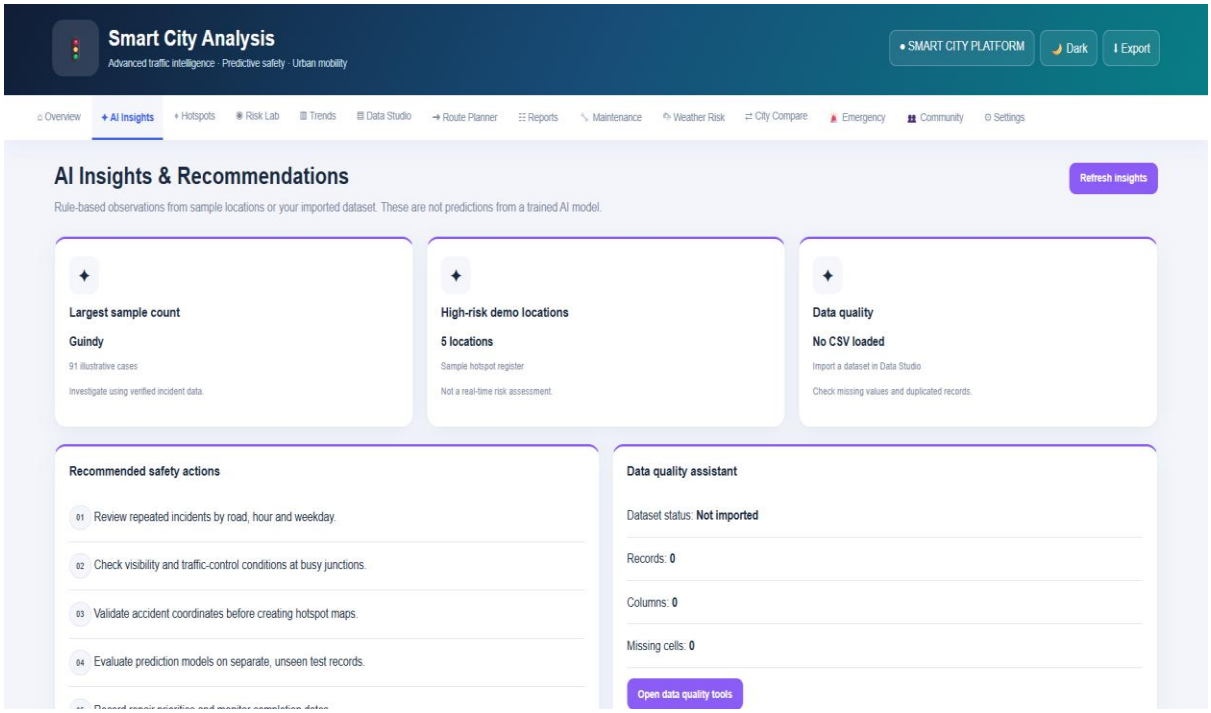
The final report should include the actual test results. An expected result is not evidence that the test passed.

CHAPTER 6 — PROJECT DEMONSTRATION

This chapter explains how to present the running dashboard during a project demonstration. Add screenshots from your actual application under the corresponding subsections. The image references below are illustrative and are not screenshots of your implemented dashboard.

6.1 Executive Overview Dashboard





The executive overview is the main entry point of the platform. It provides a consolidated summary of the available road-accident data and allows users to identify important indicators quickly.

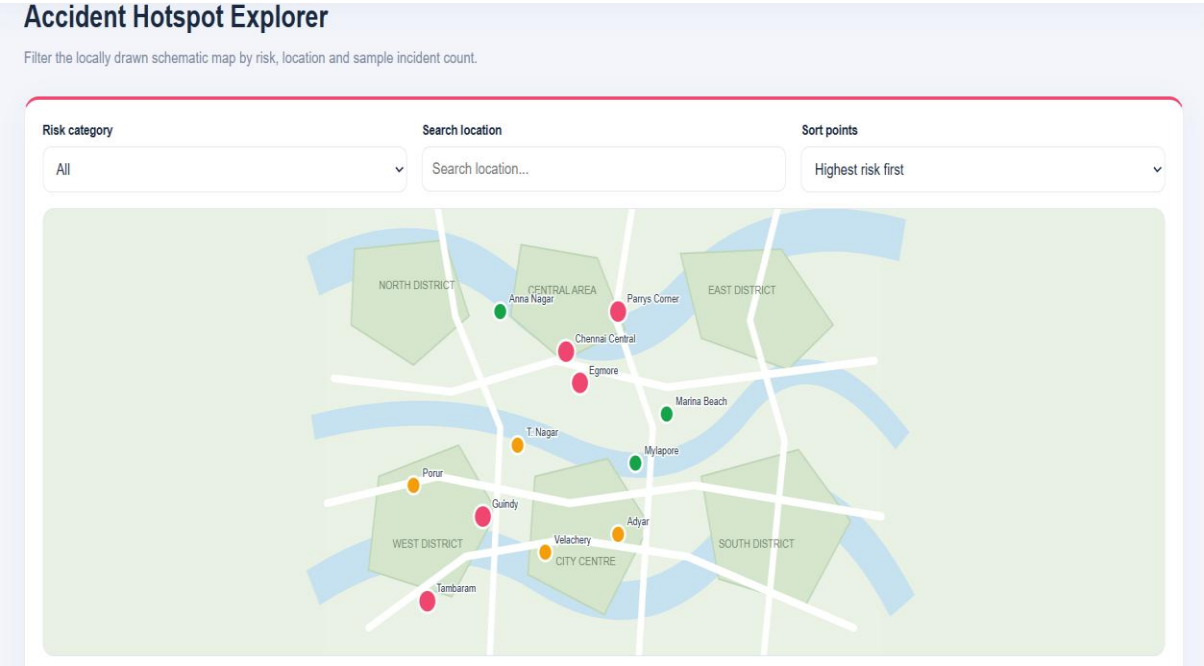
The overview can display:

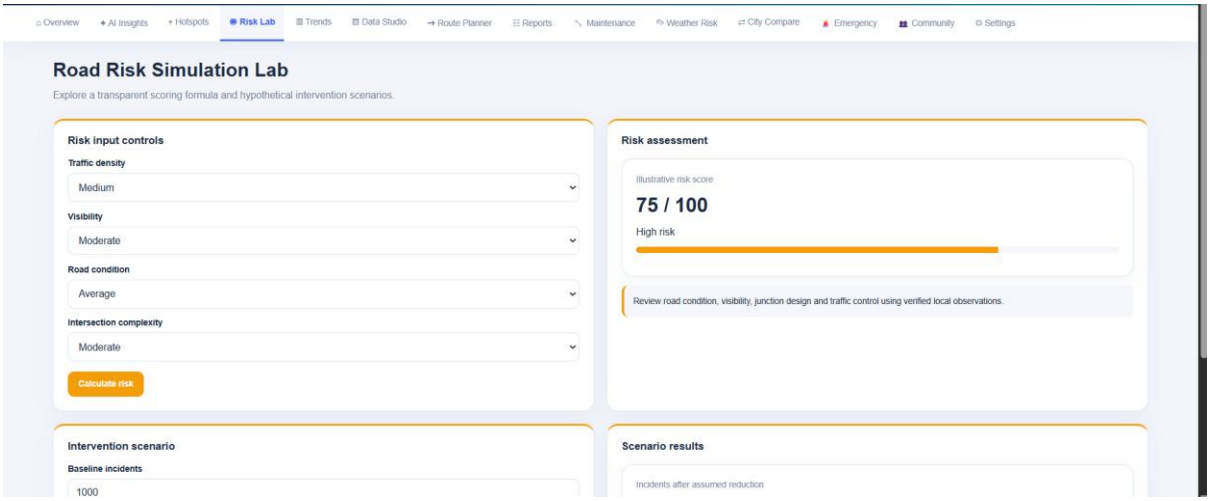
- Total recorded accidents.
- Total recorded casualties.
- Number of fatal-severity records.
- Average risk score, if the dataset defines a valid score.
- Accident severity distribution.
- Accident counts across selected cities.
- Recent or historical accident trends.

Each indicator must be calculated from the actual dataset. For example, the total accident count should be derived from the number of valid accident records or distinct accident identifiers, depending on the data structure.

The dashboard should avoid presenting illustrative values as real results.

6.2 Geographic Accident Map





The geographic accident map presents accident locations using the available geographic coordinates. It allows users to explore spatial distributions and inspect the attributes of individual records.

Potential features include:

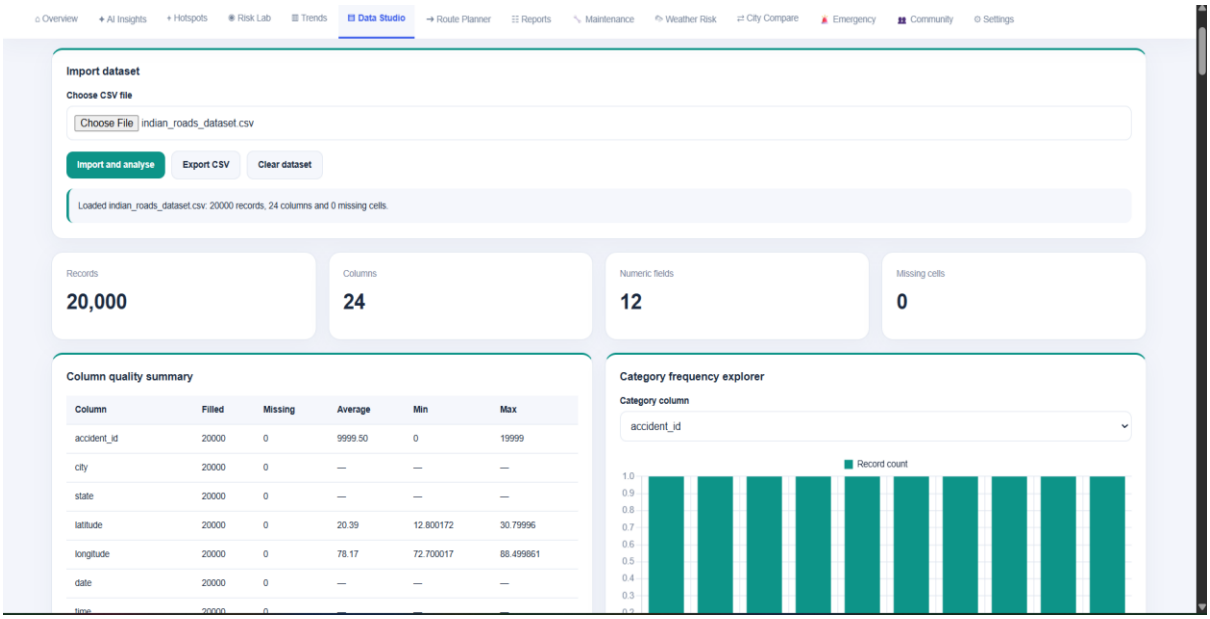
- Zooming and panning.
- Accident markers.
- Severity-based marker styles.
- Marker clustering for dense locations.

- Popups containing selected accident attributes.
- Filters for location, severity, and date.
- Heatmap visualization where appropriate.

The map can support exploratory identification of areas with many recorded incidents. A high concentration of markers does not, by itself, prove that a road is more dangerous than another road with different traffic exposure.

6.3 Accident Trend Analysis



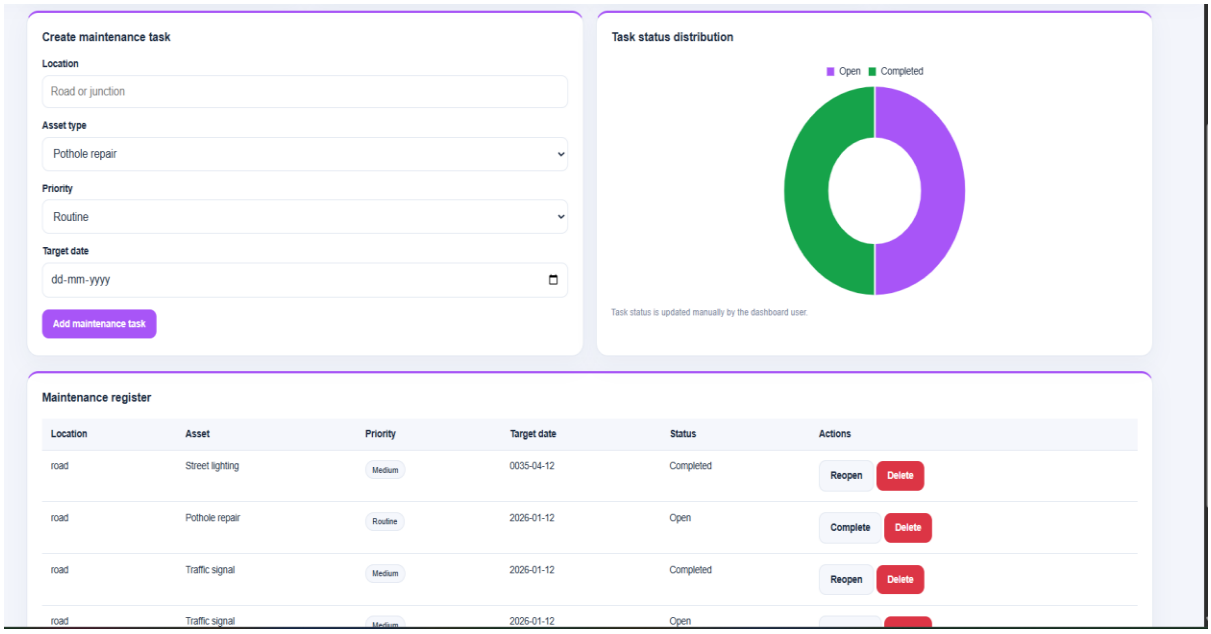


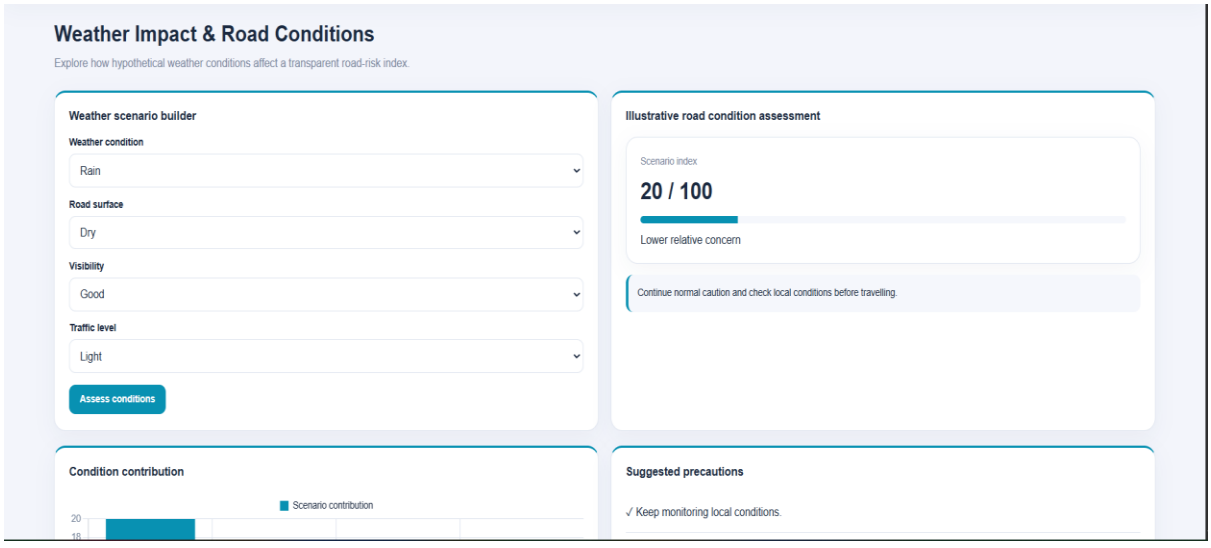
This module analyzes changes in recorded accidents across time periods.

The hourly chart compares the number of recorded accidents during different hours of the day. A weekday comparison examines differences between days of the week. Monthly and yearly charts provide a broader view of the temporal distribution.

Users can select a period or category to investigate a subset of the records. These comparisons can help identify patterns for further study, but they should not be interpreted as proof that a particular hour or day causes accidents.

6.4 Traffic and Risk Analysis





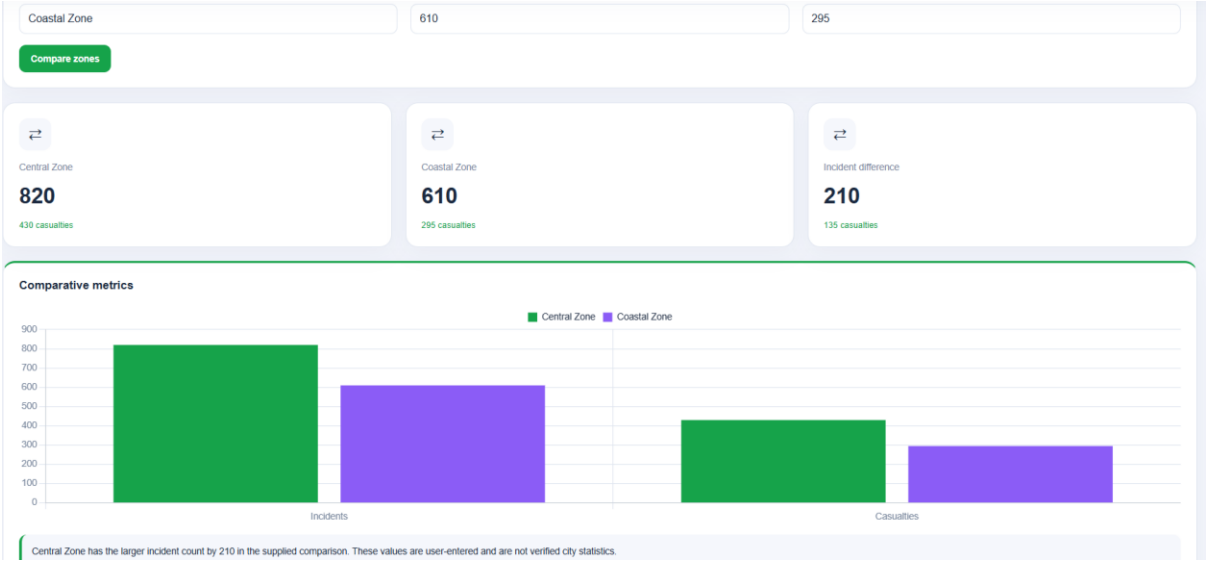
The traffic and risk analysis module compares accident records across available road and environmental conditions.

Examples include:

- Accident counts by weather category.
- Severity distribution by road type.
- Accident records by traffic density.
- Comparison of visibility categories.
- Peak-hour and non-peak-hour summaries.
- Analysis of recorded risk scores, where the score definition is documented.

This module should distinguish raw accident counts from accident rates and model-derived indicators. If a risk score is only a heuristic, its calculation and limitations should be explained beside the chart.

6.5 Machine-Learning Insights



This module presents the results generated by the accident-severity classification model.

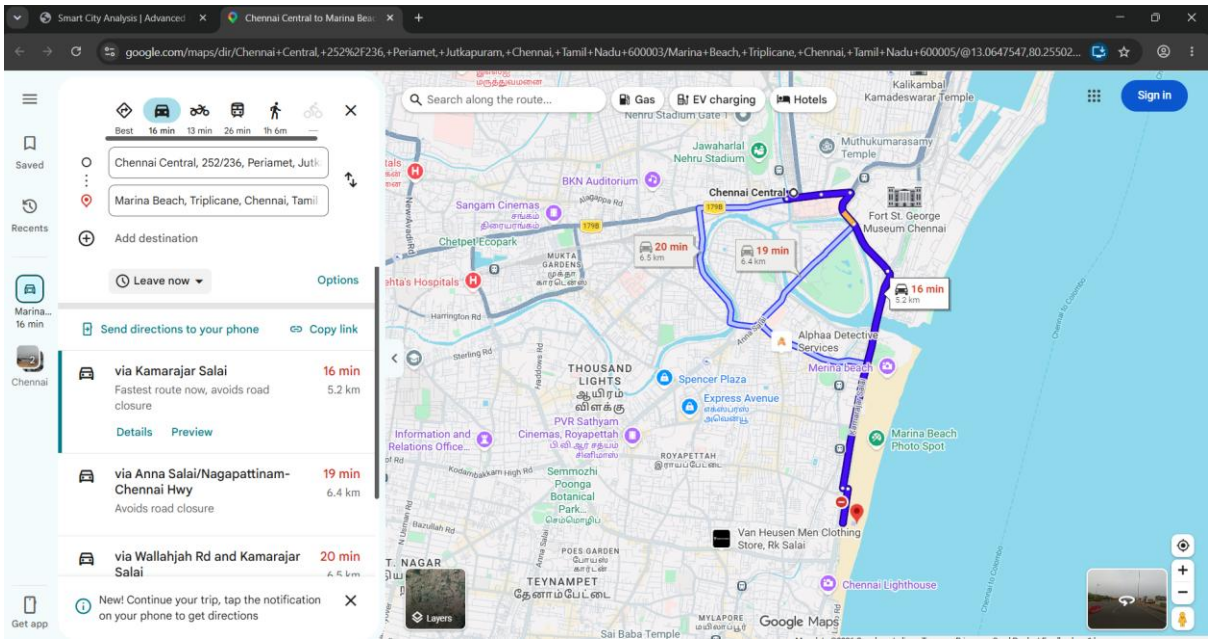
It may include:

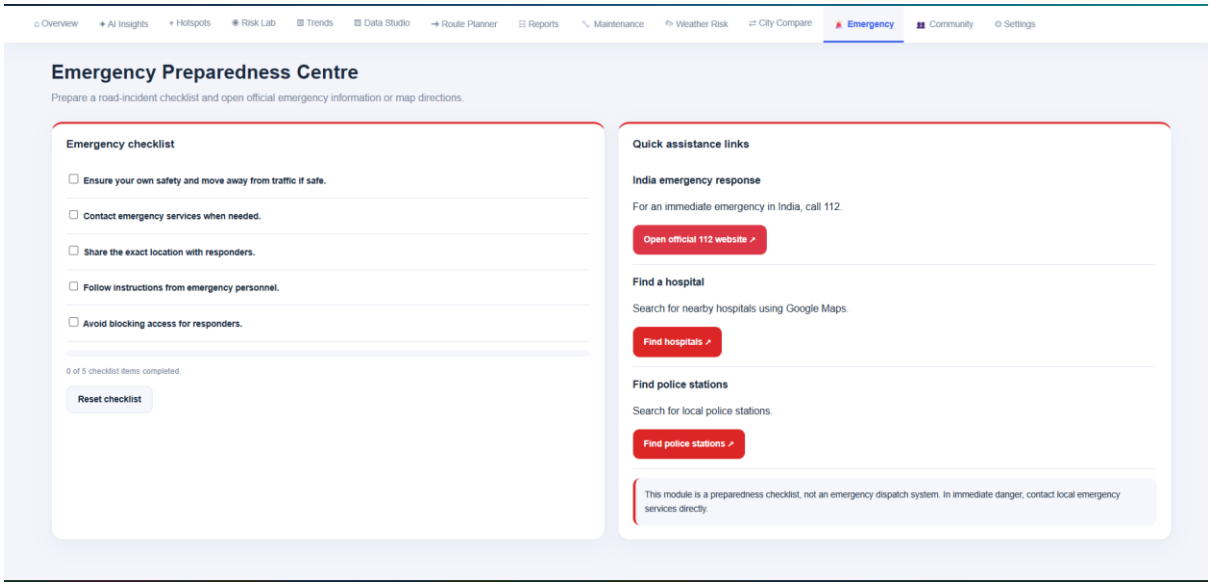
- Model name and training configuration.
- Training and test sample counts.
- Accuracy, precision, recall, and F1-score.
- A confusion matrix.
- Class-wise evaluation metrics.
- Feature-importance visualization.
- Predicted and actual severity comparisons.

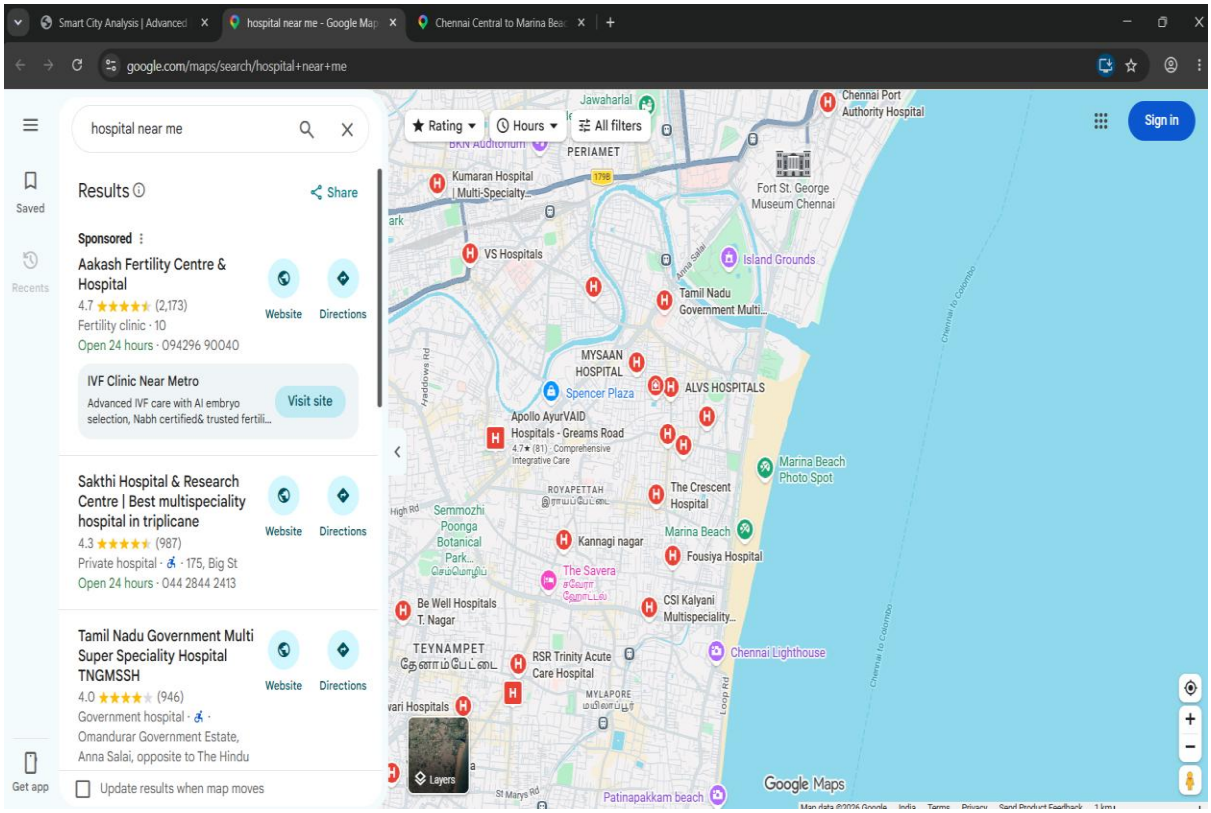
The module should clearly identify whether the displayed metrics come from training, cross-validation, or the held-out test set. Held-out evaluation is important for estimating how well the model performs on data not used to fit it.

A predicted severity category should not be presented as a guaranteed outcome. Model predictions can be wrong, especially for underrepresented categories.

6.6 Data Explorer







The data explorer allows users to inspect individual accident records and verify the information used by the analytical charts.

Possible features include:

- Search by accident identifier.
- Filter by city or state.
- Filter by severity.
- Select a date range.
- Filter by weather or road type.
- Display selected columns.
- Show the number of records matching the filters.
- Export filtered records if implemented.

The table and charts should use consistent filtering logic so that the displayed totals agree with the records being examined.

CHAPTER 7 — RESULTS AND DISCUSSION

Important: This chapter provides the reporting format and interpretation method. Actual results must be obtained by running your code against the final dataset. The existing sample report contains separate Smart Traffic results, but those results should not automatically be attributed to the updated platform.

7.1 Dataset Characteristics

The first stage of evaluation records the size and structure of the dataset used in the final experiment.

Attribute	Result to report
Total input records	[Insert actual count]
Number of columns	[Insert actual count]
Records retained after cleaning	[Insert actual count]
Missing values before cleaning	[Insert actual count or summary]
Duplicate records investigated	[Insert actual count]
Geographic coverage	[Insert verified locations]
Time period	[Insert verified date range]
Severity categories	[Insert actual categories]
Training records	[Insert actual count]
Testing records	[Insert actual count]

These values should be generated from the dataset and preprocessing outputs.

7.1.1 Severity distribution

The severity distribution chart describes how accident records are divided across the available severity categories.

Discuss the largest and smallest categories using the actual counts and percentages. If the target classes are imbalanced, explain how that affects model evaluation.

7.1.2 Geographic distribution

The geographic visualization compares the distribution of recorded accidents across the locations represented in the dataset.

Discuss the locations with the highest and lowest recorded counts. Explain that the comparison is based on the available records and may not account for differences in traffic volume, population, road length, or reporting completeness.

7.1.3 Temporal distribution

The temporal analysis examines accident counts by hour, day, month, and year, depending on the fields available.

Discuss any visible peaks or variations that appear in the actual charts. If the dataset covers multiple years, consider whether changes in reporting coverage or data collection may affect the interpretation.

7.2 Visualization Findings

The visualization results should be described in terms of patterns that are visible and supported by the underlying records.

Analysis	What to report
Severity	Counts and percentages in each category
City-wise comparison	Locations with the largest recorded counts
Weather analysis	Differences in recorded accident counts or severity distributions
Road-type analysis	Recorded distributions across road categories
Time-series analysis	Peaks, declines, and variations over time
Geographic analysis	Spatial concentrations supported by valid coordinates
Risk indicators	Definitions, comparisons, and limitations of each score

Do not invent explanations for a pattern simply because it appears in a chart. Observed differences can suggest further investigation without establishing causation.

7.3 Model Evaluation

The machine-learning model should be evaluated using a held-out test dataset.

Record the actual results in the following table.

Metric	Random Forest result
Test accuracy	[Actual value]
Macro precision	[Actual value]
Macro recall	[Actual value]
Macro F1-score	[Actual value]
Weighted F1-score	[Actual value]
Minor-class recall	[Actual value, if applicable]
Major-class recall	[Actual value, if applicable]

Fatal-class recall	[Actual value, if applicable]

If any of the categories do not exist in your actual dataset, remove those rows or replace them with the correct category names.

7.3.1 Confusion matrix analysis

The confusion matrix shows how often each actual category was assigned to each predicted category.

Use it to identify:

- Classes that are frequently classified correctly.
- Classes that are often confused with other categories.
- Underrepresented classes with poor recall.
- Possible class imbalance or model limitations.

A high overall accuracy does not necessarily mean that the classifier performs well on every severity category.

7.3.2 Feature-importance analysis

The feature-importance chart identifies the variables that contribute most according to the selected model-importance measure.

Discuss the highest-ranked features from the actual model output. Explain that feature importance can reflect correlations, dataset structure, and model behavior; it does not independently establish that a feature causes an accident.

7.3.3 Comparison with a baseline

If possible, compare Random Forest with a simple baseline classifier or another suitable model. Both models should use the same evaluation split and comparable preprocessing.

Evaluation aspect	**Baseline model**	**Random Forest**
Test accuracy	[Actual result]	[Actual result]
Macro F1-score	[Actual result]	[Actual result]
Minority-class recall	[Actual result]	[Actual result]
Training time	[Measured time]	[Measured time]

This comparison helps determine whether the additional complexity of Random Forest provides a meaningful improvement.

7.4 Limitations

The following limitations should be considered:

- Dataset coverage: The dataset may not represent all cities, roads, or time periods.
- Missing data: Incomplete fields can affect descriptive summaries and classification.
- Class imbalance: Some severity categories may have too few examples for reliable classification.
- Model generalization: Test performance may not transfer to locations or periods outside the training data.
- Geographic accuracy: Inaccurate coordinates can misrepresent accident locations.
- Causal interpretation: Observed associations do not prove that a particular factor causes accidents.
- Risk calibration: A dataset-derived score is not automatically a calibrated probability of a future accident.
- Operational deployment: A local academic prototype is not necessarily suitable for real-time traffic control or emergency response.

CHAPTER 8 — CONCLUSION AND FUTURE WORK

8.1 Conclusion

The Smart City Traffic and Accident Analytics Platform Using Advanced Interactive Data Visualization and Machine Learning is designed to improve the accessibility and interpretation of historical road-accident information.

The project combines data preparation, exploratory analysis, interactive visualization, geographic mapping, and machine-learning classification within a unified analytical workflow. It provides a structured method for investigating accident severity, location-based differences, temporal patterns, weather conditions, road categories, and other attributes available in the dataset.

The interactive dashboard allows users to move from summary-level indicators to more detailed charts and individual records. The machine-learning component provides an

additional approach to studying severity categories, while its evaluation metrics and confusion matrix help communicate the strengths and weaknesses of the trained model.

The primary contribution is the integration of these components into a single analytical platform. The project demonstrates how data visualization and machine learning can complement each other when exploring transportation data.

The reliability of the results depends on data quality, correct preprocessing, reproducible model evaluation, and careful interpretation. The current system should be regarded as an analytical prototype rather than a validated real-time accident-prevention system.

8.2 Future Work

The platform can be improved in several directions.

1. Real-time data integration: Connect verified live traffic or incident feeds to update the dashboard as new information becomes available.

2. Improved prediction models: Evaluate alternative classifiers, tune model parameters, and investigate techniques for handling class imbalance.

3. More robust evaluation: Use cross-validation, class-wise metrics, calibration checks, and suitable time-based or location-based validation where appropriate.

4. Advanced geographic analysis: Incorporate verified road-network data and more detailed geographic aggregation. Where exposure information is available, investigate normalized accident rates.

5. Explainable machine learning: Add model-interpretation methods that help users understand individual predictions and the influence of input variables.

6. Interactive route analysis: Explore integration with a reliable road-network or routing service, while distinguishing route planning from accident prediction.

7. Automated reporting: Generate downloadable summaries containing selected charts, tables, and model metrics.

8. Accessibility improvements: Improve keyboard navigation, color contrast, responsive layouts, and chart descriptions.

9. Privacy and security: Apply appropriate access controls, minimize unnecessary personal information, and document data-retention practices.

10. Deployment and scalability: Test the platform under larger datasets and, if required, deploy it to a secured server with suitable performance monitoring.

These improvements would help develop the prototype into a more comprehensive transportation analytics system, subject to appropriate data access and validation.

CHAPTER 9 — REFERENCES

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Paper Name: A Review on Action Recognition for Accident Detection in Smart City Transportation Systems
Link: https://link.springer.com/content/pdf/10.1186/s43067-023-00124-y.pdf

2. Author(s): Muhammad Shujaat Abid et al.
Paper Name: Improving Road Safety in Smart Cities Using Machine Learning Techniques
Link: https://www.nature.com/articles/s41598-026-36795-6

3. Author(s): Ayman Mohamed Mostafa et al.
Paper Name: AI-Based Prediction of Traffic Crash Severity for Improving Road Safety and Transportation Efficiency
Link: https://www.nature.com/articles/s41598-025-10970-7

4. Author(s): Humera Khanum, Anshul Garg, Mir Iqbal Faheem, and Rushikesh Kulkarni
Paper Name: A Methodological Framework for Road Accident Severity Prediction for Indian Highways Using Machine Learning Models
Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC12686657/

5. Author(s): Izuchukwu Chukwuma Obasi and Chizubem Benson
Paper Name: Evaluating the Effectiveness of Machine Learning Techniques in Forecasting the Severity of Traffic Accidents
Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC10407198/

6. Author(s): Yetay Berhanu, Dietrich Schröder, Bikila Teklu Wodajo, and Esayas Alemayehu
Paper Name: Machine Learning for Predictions of Road Traffic Accidents and Spatial Network Analysis for Safe Routing on Accident and Congestion-Prone Road Networks
Link: https://www.sciencedirect.com/science/article/pii/S2590123024009927

7. Author(s): Authors listed in the publisher's paper record
Paper Name: Enhancing Road Safety with Machine Learning: Current Advances and Future Directions in Accident Prediction Using Non-Visual Data
Link: https://www.sciencedirect.com/science/article/pii/S0952197624012442

8. Author(s): Authors listed in the publisher's paper record
Paper Name: A Study on Road Accident Prediction and Contributing Factors Using Explainable Machine Learning Models: Analysis and Performance
Link: https://www.sciencedirect.com/science/article/pii/S2590198223000611

9. Author(s): Pengfei Cui, Xiaobao Yang, Mohamed Abdel-Aty, Jinlei Zhang, and Xuedong Yan
Paper Name: Advancing Urban Traffic Accident Forecasting Through Sparse Spatio-

Temporal Dynamic Learning
Link: https://www.sciencedirect.com/science/article/pii/S000145752400109X

10. Author(s): Kun Li, Kaiyang Guo, Zhantu Liang, Haocheng Xu, and Xinlong Duan
Paper Name: Traffic Accident Severity Prediction and Analysis via Spatio-Temporal Deep Learning: A ConvLSTM-Transformer Approach
Link: https://www.sciencedirect.com/science/article/pii/S0960077925010604